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BCA(Sem-VI)-Artificial

Intelligence (601) Core-XIII

2024

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer from all the section as directed.

Section-A

Multiple choices questions

(Compulsory)

1. Choose the correct answer from the following
multiple choice questions : $2 \times 10 = 20$

(i) What is the primary objective of artificial
Intelligence ?

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(1)

(Turn over)

- (a) To create machines that can think and reason like humans
 - (b) To develop software that can automate tasks
 - (c) To make human life easier
 - (d) To create intelligent agents that perceive and act in the environment
- (ii) What is the primary objective of Machine Learning in AI ?

- (a) To store large dataset
- (b) To enable machines to learn from data and improve over time without being explicitly programmed
- (c) To enhance computer graphics
- (d) To perform predefined tasks without adjustments

(iii) What is the purpose of the Turing Test in AI?

Contd.

- (a) To evaluate the speed of an algorithm
 - (b) To determine whether a machine can exhibit intelligent behaviour indistinguishable from a human.
 - (c) To assess the accuracy of an AI system's responses
 - (d) To test the ethical boundaries of AI usage
- (iv) Which of the following best describes a "rational agent"?
- (a) that acts randomly in the environment
 - (b) that acts to maximize its performance measure
 - (c) that always follows predefined rules
 - (d) that mimics human emotions and behaviours
- (v) In AI, which of the following environments is classified as "deterministic" ?

- (a) A card game like Poker
 - (b) A robot vacuum cleaner's environment
 - (c) Chess
 - (d) A video game with random obstacles
- (vi) What is an intelligent agent ?

- (a) A software program that can solve complex mathematical problems
- (b) An entity that perceives its environment and take actions to maximize its chances of success
- (c) A machine that mimics human emotions and behaviours
- (d) A robotic system designed for factory automation

(vii) What does PEAS stand for in the context of Intelligent ?

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(4)

Contd.



- (a) Performance, Environment,
Actuators, Sensors
- (b) Perception, Execution, Action,
Success
- (c) Programming, execution, Analysis,
Strategy
- (d) Planning estimation, application,
simulation

(viii) Means End Analysis is a problem solving strategy that involves :

- (a) Random exploration of solutions
- (b) Directly comparing the current
state and goals state to reduce the
difference
- (c) Exhaustively searching all possible
solutions.
- (d) Avoiding the use of heuristics

(ix) Which of the following methods is commonly used in Constraint Satisfaction Problems (CSP) to reduce the search space ?

- (a) Backtracking
- (b) Forward checking
- (c) Depth First Search
- (d) Hill Climbing

(x) In the context of Hill Climbing, which of the following is a common issue :

- (a) Excessive memory usage
- (b) Always finding the global optimum
- (c) Running out of nodes to explore
- (d) Getting stuck in local maxima

Section-B

Short answer type questions

Answer any **four** questions of the following :

$$5 \times 4 = 20$$

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(6)

Contd.



2. Define Semantic Network.
3. How Logical Connectives are used in Propositional Logic ?
4. What is FOPL? Where can it be used in AI ?
5. What are the disadvantages of AI system ?
6. Write short notes on "Expert System".
7. Differentiate between DFS and BFS.
8. What are problem characteristics?

Section-C

Long answer type questions

Answer any **two** questions of the following :

15×2=30

9. Explain in details about AI agents with their diagrams.
10. What do you mean by Game Playing in AI ?
What are its types? Also explain the techniques used in Game playing.

11. How do we deal with Uncertainties and Inconsistencies in AI ? Explain the common approaches.

12. Write short notes on any **three** of the following :

- (a) Rational Agent Approach
- (b) Turing Test
- (c) PROLOG
- (d) Resolution and unification
- (e) Knowledge Representation in AI
- (d) Conceptual Dependencies

—————x—————